

2601-MTL106/MTL2006: TUTORIAL-3

Q1. An airline estimates that 5% of the passengers who make reservations for a particular flight will not show up. To account for these no-shows, the airline sells 55 tickets for a flight that has only 53 seats. Assume that each passenger independently decides whether or not to show up. What is the probability that there will be a seat available for every passenger who arrives at the airport?

Q2. The probability that a single shot hits an aircraft is 0.05. Assume that the outcomes of different shots are independent, so that the number of hits in n shots follows a binomial distribution. How many shots must be fired so that the probability of obtaining at least two hits exceeds 0.95?

Q3. For any $X \sim \mathcal{B}(n, p)$, show that

$$\mathbb{E}\left[\frac{1}{1+X}\right] = \frac{1 - (1-p)^{n+1}}{(n+1)p}.$$

Q4. A communication system consists of n independent components. Each component functions correctly with probability p . The system is considered to operate effectively if at least half of its components are functioning. For what values of p is a 5-component system more likely to operate effectively than a 3-component system?

Q5. Let $X \sim \mathcal{P}(\lambda)$ be a Poisson random variable with parameter λ . Prove the following:

- a) For every positive integer n , $\mathbb{E}[X^n] = \lambda \mathbb{E}[(1+X)^{n-1}]$.
- b) Show that $\mathbb{P}(X \text{ is even}) = \frac{1}{2}(1 + e^{-2\lambda})$ and $\mathbb{P}(X \text{ is odd}) = \frac{1}{2}(1 - e^{-2\lambda})$.

Q6. Suppose that the average number of accidents occurring each week on a particular highway is 5. Assuming that the number of accidents in a week follows a Poisson distribution, calculate the probability that at least two accidents occur during a given week.

Q7. A manufacturer produces light bulbs that are packaged in boxes of 1000. Quality control studies indicate that 0.5% of the light bulbs produced are defective. Determine the percentage of boxes that contain:

- (a) no defective light bulbs,
- (b) two or more defective light bulbs.

Q8. A large batch of automobile tires contains 5% defective tires. Six tires are selected one at a time to equip a car. The selection process continues until 6 good tires have been found.

- i) Find the probability that 3 defective tires are found before the 6th good tire.
- ii) Find the mean and variance of the number of defective tires found before the 6th good tire.

Q9. Let $X \sim \mathcal{B}_N(r, p)$ denote a negative binomial random variable representing the number of failures before the r -th success, and let $Y \sim \mathcal{B}(n, p)$ denote a binomial random variable with parameters n and p . Show that

$$\mathbb{P}(X > n) = \mathbb{P}(Y < r).$$

Q10. A stick of length 1 is broken at a point U , where U is uniformly distributed over the interval $(0, 1)$. Let L denote the length of the piece that contains the point $\frac{1}{3}$. Determine $\mathbb{E}(L)$ and $\text{Var}(L)$.

Q11. A point is selected at random from the interval $[-1, 3]$, with all points being equally likely. If X denotes the selected point and $Y = X^2$, determine the probability density function of Y .

Q12. Suppose X and Y are random variables, where X denotes the speed and $Y = \frac{180}{X}$ denotes the travel time (duration). If X is uniformly distributed on the interval $[30, 60]$, find the probability density function (pdf) of Y .

Q13. Consider the marks obtained by students in the Mathematics examination. Assume that the marks are normally distributed with a mean of 56 and a standard deviation of 18. It is observed that 10% of the students receive an A grade, while 5% of the students fail the course.

- i) Determine the minimum mark required to obtain an A grade.
- ii) Determine the minimum mark required to pass the course.

Q14. A manufacturer produces bolts whose diameters are normally distributed with a mean of 1.2 inches and a standard deviation of 0.01 inches. To satisfy quality specifications, each bolt must have a diameter between 1.19 inches and 1.21 inches. What percentage of the bolts produced do not meet these specifications?

Q15. Let $f(x)$ be the probability density function of a normally distributed random variable with mean μ and variance σ^2 . Show that the points $x = \mu - \sigma$ and $x = \mu + \sigma$ are points of inflection of the graph of $f(x)$. In other words, prove that $f''(x) = 0$ when $x = \mu - \sigma$ and $x = \mu + \sigma$.

Q16. Let X be a normally distributed random variable with mean μ and variance σ^2 . If $Y = e^X$, determine the probability density function of Y .

Q17. The time (in hours) required to repair a machine is an exponentially distributed random variable with parameter $\lambda = 1$.

- i) Find the probability that the repair time exceeds 2 hours.
- ii) Given that the repair has already lasted more than 2 hours, find the probability that it requires at least 3 hours to complete.

Q18. The lifetime (in years) of a radio is exponentially distributed with a mean of 8 years. If Jones purchases a used radio, what is the probability that it will continue to function for at least another 8 years?

Q19. The time to failure T (in operating hours) of a compressor is a continuous random variable with probability density function

$$f(t) = \begin{cases} \frac{0.002}{(0.001t+1)^3}, & t \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

Determine the following:

- i) Find the failure rate (hazard) function of the compressor.
- ii) Determine the reliability of the compressor for an operating life of 100 hours.
- iii) Determine the design life of the compressor if a reliability of 0.95 is required.
- iv) Determine the mean time to failure (MTTF) of the compressor.

Q20. Let X be a uniformly distributed random variable on the interval $(0, 2)$. Determine the failure rate (hazard) function of X . If $Y = 3X$, find the failure rate (hazard) function of Y .

Q21. The time (in hours) taken by a customer to choose a car at a Honda showroom is gamma distributed with a mean of 2 hours and a variance of 2 hours².

- i) What is the probability that a customer takes no more than 4.5 hours to choose a car?
- ii) Find the 90th percentile of the time required to choose a car.

Q22. Prove that the reciprocal of a standard Cauchy random variable is also a standard Cauchy random variable.

Q23. Let X be a random variable having probability density function

$$f(x) = kx^{-\frac{1}{2}}(1-x)^{\frac{1}{2}}, \quad 0 < x < 1,$$

where k is a normalizing constant. Determine the value of k . Find the mean and variance of X .

Q24. Determine whether the continuous uniform distribution is a special case of the beta distribution. If so, identify the corresponding values of the parameters.